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Functional Magnetic Resonance Imaging

Third Edition

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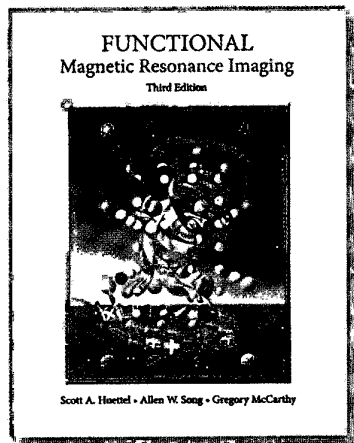
Scott A. Huettel
Duke University

Allen W. Song
Duke University

Gregory McCarthy
Yale University



Sinauer Associates, Inc. *Publishers*
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An Introduction to fMRI

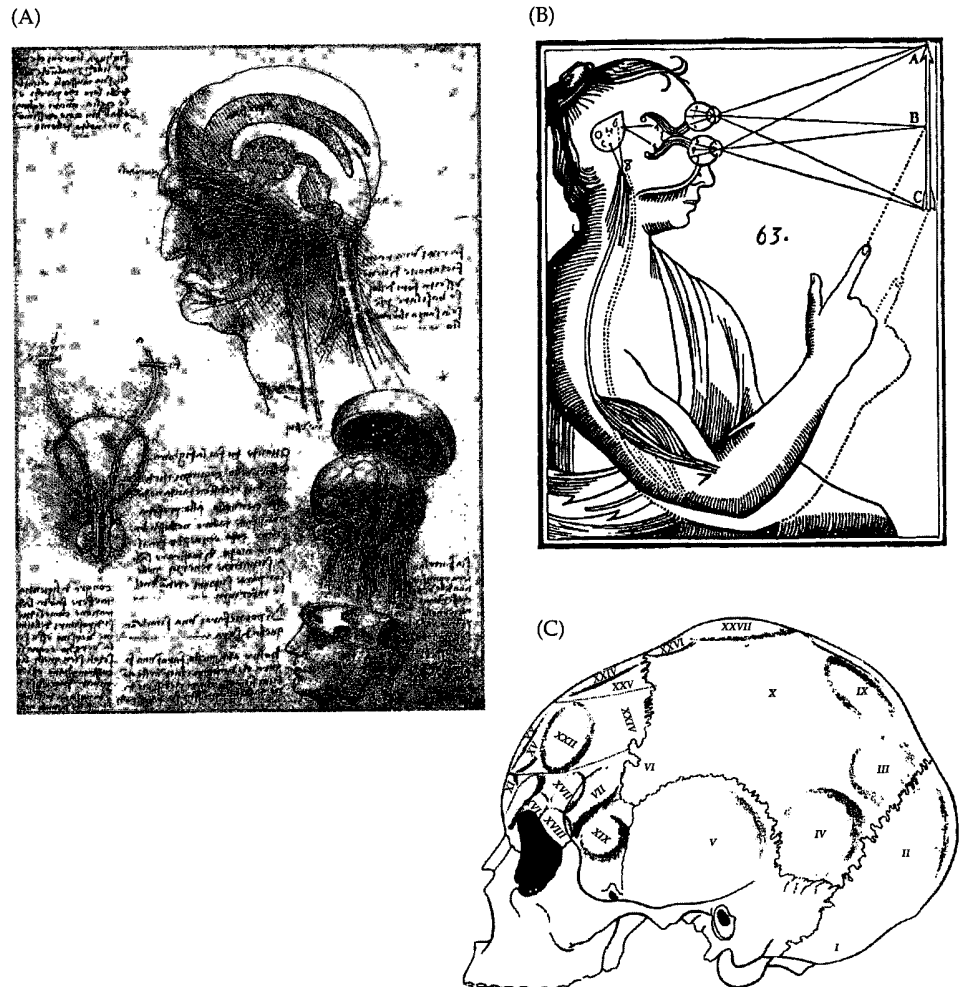
Few scientific developments have been more striking than the ability to make images of the functioning human brain. Why do these images evoke such wonder? To many, the human brain represents a barely explored new world, with each image providing a glimpse of hidden structures. Like the navigational charts used by medieval explorers, current maps of brain function are riddled with errors, inconsistencies, and puzzles deserving of solution. Yet the difficulty in understanding the brain has added to the excitement of the quest.

Renaissance and post-Renaissance physicians and philosophers developed only very rough theories about brain function (**Figure 1.1A,B**). The philosopher Rene Descartes speculated in the mid-seventeenth century that mental activity arose through the influence of the pineal gland on the surrounding ventricles. Others of his contemporaries argued that the corpus callosum, whose fibers connect the two hemispheres of the brain, provided the basis for all cognition. Perhaps most prescient of all was Emanuel Swedenborg, who began his career as an engineer and ended as a theologian. Writing in the mid-eighteenth century, Swedenborg postulated that the cortex itself was responsible for higher cognition, and that different parts of the cortex were associated with different functions. He argued that the frontal lobes were associated with imagination, thought, and will; this was a surprisingly accurate description, given the complete lack of experimental evidence at that time. But Swedenborg's ideas were completely ignored by his contemporaries—only to be rediscovered two centuries later, as a historical curiosity.

By the nineteenth century, scientists began to systematically pursue the idea that different aspects of the mind were represented in different brain regions, or *localization of function*. Some of their attempts involved now-discredited methods, like examining bumps on the skull ("phrenology") to draw inferences about the underlying cortical volume. The resulting maps (**Figure 1.1C**) were drawn largely from anecdotes (e.g., observations of individuals with extreme characteristics) and were not tested using experimentation. Thus, they were eventually abandoned because of a lack of scientific grounding. Better evidence came from cases of brain damage, as naturally occurred in humans or created in laboratory studies in animals. Such studies provided

localization of function The idea that the brain may have distinct regions that support particular mental processes.

Figure 1.1 Some early views of brain function. (A) While the Italian genius Leonardo da Vinci is most famous for his paintings and inventions, he also had boundless curiosity about the natural world, including the human form. His brain dissections included injection of hot wax into the ventricular cavities, which allowed him to make some of the earliest accurate maps of those fluid-filled structures. Consistent with the ideas of many of his contemporaries, he mapped different functions onto different parts of the ventricular system (drawings c. 1508). (B) Descartes likewise believed that the ventricles were critical for brain function, in that they contained psychic “spirits” that flowed from the pineal gland (shown here in a drawing from 1662). In his framework, the single pineal gland was the seat of the mind. (C) The phrenological mapping system created by Franz Joseph Gall (c. 1810). The phrenologists believed that people with an extreme trait (e.g., very wise; prone to thievery) would have an abundance of cortex devoted to that function. To find out which brain area was associated with the trait, researchers would examine the skulls of such people for bumps or protrusions. Each numbered region in this figure represents a different trait, from “reproductive instinct” (I) to “firmness of purpose” (XXVII).



critical information about the gross organization of the brain, which in turn led to important insights about the nature and treatment of many neurological disorders. Yet, animal models and naturally occurring brain damage were insufficient for studying some complex brain functions, and direct manipulation or invasive measurement of the human brain were prohibited on ethical grounds.

Today, a new group of explorers are mapping the human brain, often using techniques that allow the noninvasive measurement of the human brain as it performs complex functions. Many of these scientists use functional magnetic resonance imaging, or fMRI, to measure the active brain in both clinical and research settings. Localizing functions within the brain remains a key goal for the field; by using careful experiments that isolate particular functions, scientists build maps of how the brain is organized. In recent years, fMRI has increasingly been used to understand brain connectivity (in the mapping metaphor, routes between distinct locations) through measurements of how sets of brain regions interact to shape changes in function over time. And, concurrent with the growth of basic science using fMRI, there has been a similar growth in applications of fMRI to understand individual differences

functional magnetic resonance imaging (fMRI) A neuroimaging technique that uses standard MRI scanners to investigate changes in brain function over time.

in personality, the effects of genetic differences on brain function, and the consequences of neurological and psychiatric disorders. In barely 20 years, fMRI has grown from a theoretical concept to become the dominant technique in cognitive neuroscience.

What Is fMRI?

As its name implies, magnetic resonance imaging (MRI) uses strong magnetic fields to create images of biological tissue. The strength of the static magnetic field created by an MRI scanner is expressed in units of tesla (T) (one tesla is equal to 10,000 gauss). A typical scanner used for fMRI today (**Figure 1.2**) has a field strength of 3.0 tesla (3.0 T), but field strength can range from 1.5 T to 7.0 T or even higher. For comparison, the strength of Earth's magnetic field is approximately 0.00005 T.

To create images, the scanner uses a series of changing magnetic gradients and oscillating electromagnetic fields, known as a **pulse sequence**. Depending on the frequency of the electromagnetic fields, energy may be absorbed by atomic nuclei. For MRI, scanners are tuned to the frequency of hydrogen nuclei, which are the most common in the human body due to their prevalence in water molecules. After it is absorbed, the electromagnetic energy is later released by the nuclei. The amount of released energy depends on the numbers and types of nuclei present.

Depending on the pulse sequence used, an MRI scanner can detect different tissue properties and distinguish between tissue types. For example, an MRI of the knee can reveal whether ligaments are intact or torn, and an MRI of the brain can detect the difference between gray and white matter. Different pulse sequences can be constructed that create images sensitive to tumors, abnormalities in blood vessels, bone damage, and many other conditions.

static magnetic field The strong magnetic field at the center of the MRI scanner whose strength does not change over time. The strength of the static magnetic field is expressed in units of tesla (T).

pulse sequence A series of changing magnetic field gradients and electromagnetic pulses that allows the MRI scanner to create images sensitive to a particular physical property.

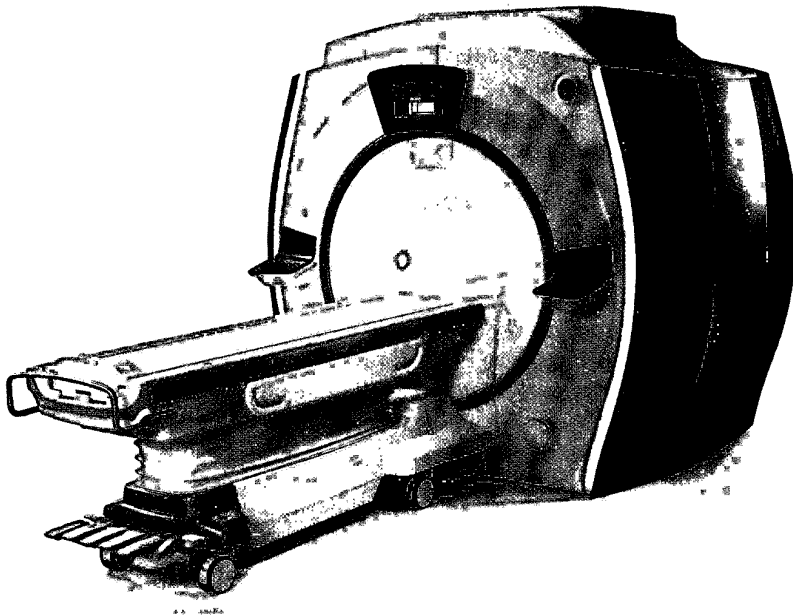


Figure 1.2 A modern MRI scanner. The main magnetic field of the scanner shown is 3.0 Tesla, or about 60,000 times the strength of Earth's magnetic field. The subject lies down on the table at the front of the scanner; depending on the imaging protocol, there may be additional equipment surrounding the body part to be scanned (e.g., a head coil). The table then moves back into the bore of the scanner until the targeted body part is positioned at the scanner's center. (Courtesy of GE Healthcare, Waukesha, WI.)

structural neuroimaging A class of research and clinical techniques that create images of the brain's physical structure, often to provide insight into the locations and distribution of different types of tissue.

functional neuroimaging A class of research techniques that create images of the brain's functional properties, notably different aspects of cognition and related information processing. Common functional neuroimaging techniques include fMRI, PET, and optical imaging.

positron emission tomography (PET) A functional neuroimaging technique that creates images based on the movement of injected radioactive material.

The ability to examine multiple properties of biological tissue makes MRI an extraordinarily flexible and powerful clinical tool.

Much of our knowledge about brain function has come from the study of its structure—often using **structural neuroimaging**—notably by relating neurological disorders to the patterns of brain injury that cause them. Structural studies are limited, however, in that they cannot reveal short-term physiological changes associated with the active functioning of the brain. Functional neuroimaging studies can help overcome this limitation, both by identifying specific parts of the brain where particular mental processes occur and by characterizing the patterns of brain activation associated with those processes. In effect, successful functional neuroimaging studies map patterns of brain activation to mental function. However, unlike the phrenologists, who believed that complex behaviors or personality traits were associated with discrete brain regions, modern researchers recognize that many functions rely on distributed networks, and that a single brain region may contribute to many different observed behaviors.

The first functional neuroimaging technique in common use was **positron emission tomography (PET)**, which relies on the injection of radioactive tracers to measure changes in the brain, such as blood flow or glucose metabolism. Using PET, researchers can identify the parts of the brain that are metabolically associated with a given perceptual, motor, or cognitive function, like seeing faces, moving the right hand, or mentally reciting sentences. PET imaging suffers from several disadvantages, however, including the invasiveness of the radioactive injections, the expense of generating radioactive isotopes, and the slow speed at which images are acquired. As will be discussed in Chapter 7, these factors have limited the growth of PET for many sorts of research questions, although it still has important uses. For example, PET can be used to target specific chemicals or metabolites, like neurotransmitters, and to relate the concentration of those substances to behavior, genetics, or differences among individuals.

More recently, the development of fMRI has resulted in an explosion of interest in functional neuroimaging (**Box 1.1**). Most fMRI studies measure changes in blood oxygenation over time. Because blood oxygenation levels change rapidly following the activity of neurons in a brain region, fMRI allows researchers to localize brain activity on a second-by-second basis, and within millimeters of its origin. And, because changes in blood oxygenation occur endogenously as part of normal brain physiology, fMRI is a noninvasive technique that can be repeated as many times as needed in the same individual. Because of these advantages, fMRI has been rapidly adopted as a primary investigative tool by thousands of researchers at hundreds of institutions.

Measurement versus manipulation techniques

Methods in neuroscience can be divided roughly into two classes based on how they relate brain function to behavior (**Figure 1.3**). One class, which we will call *measurement techniques*, involves the use of specialized equipment to measure brain function while subjects engage in one or more behaviors (e.g., making a motor movement, perceiving stimuli, or performing a specific form of thinking). Another class of methods, which we will call *manipulation techniques*, relies on changing the brain's structure or functioning and then examining the effects of those changes on behavior. Each has advantages and disadvantages, and many successful neuroscience research programs use both classes of techniques.

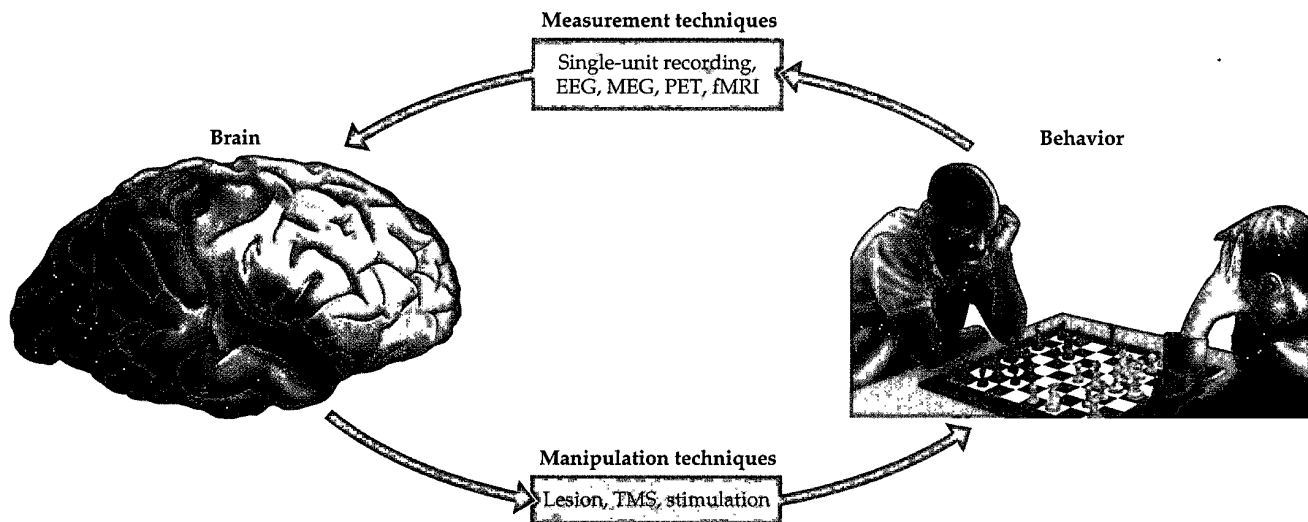


Figure 1.3 Manipulation techniques versus measurement techniques. Neuroscience techniques can be separated into two classes depending on whether they measure or manipulate brain function. When using a measurement technique, the researcher causes the subject to engage in some behavior and then measures how that behavior changes the physiological functioning of the brain. Functional MRI is a measurement technique: subjects engage in some experiment within the MRI scanner while the experimenters measure changes in blood oxygenation within the brain. Manipulation techniques adopt the reverse approach. The researcher manipulates brain function, either directly (e.g., using transcranial magnetic stimulation, or TMS) or indirectly (e.g., by comparing patients with and without brain lesions) and then examines the effect of those manipulations on behavior.

Functional MRI is a measurement technique: researchers measure changes in brain function while a subject performs an experimental task. The measurements provided by fMRI can include maps of how a particular function is represented within the brain, graphs of the relative timing of activation within a particular part of the brain, or even network diagrams that show functional relations among many different regions. Some of these types of measurements are introduced in the next section; others are described in later chapters. As noted briefly in the previous section and covered in much more detail in Chapter 3, the pulse sequences used in fMRI alter the quantum properties of individual atomic nuclei within the brain. These alterations have no effect on macroscopic functioning—they do not influence neuronal firing or blood flow.

It is critical to recognize that measurement techniques differ in what they assess. The measures collected by fMRI and PET, for example, usually provide information about brain metabolism: how much energy is being used by a particular part of the brain, as indicated by changes in oxygen (fMRI, PET) or glucose (PET) levels. Information about metabolism can be used as an indirect indicator of the local neuronal activity. Other techniques measure neuronal activity more directly. By placing an electrode near or within a single neuron, a researcher can record changes in electrical potential associated with the firing of that neuron. For example, if a monkey is trained to remember a picture over a delay of a few seconds, an electrode in one of its lateral frontal lobes will record increased activity during that delay interval. One cannot implant electrodes into healthy human subjects, although this is sometimes done in

BOX 1.1 What Is fMRI Used For?

Over the past two decades, the pace of change in fMRI research has been nothing short of dizzying. Each new discovery has generated more research questions. Each new approach for designing experiments has led to more complex paradigms and to a broader range of topics. And, each new analysis technique has enabled fMRI researchers to extract more subtle information about brain function. Progress has sparked more progress, to the point that the cutting-edge studies of today would be essentially unrecognizable to even the most prescient researcher of the 1990s.

To illustrate this progress, we here highlight a temporal series of fMRI studies that all address one of the central questions in cognitive neuroscience: the organization of visual perception. We have selected notable studies from four points in time: 1992, at the very beginning of fMRI; 1998, when researchers used more complex experimental designs and analysis approaches; 2006, with fMRI now a mature technique; and 2012, when advanced computational methods from other disciplines opened up new frontiers for data analysis. In each case, the focus is on a specific, representative result that provides insight into the cutting edge of fMRI research at that date.

1992

At the time of the first fMRI studies, much was known about brain circuitry for vision. Scientists had long recognized that lesions to the occipital and inferior temporal cortices led to a variety of deficits, from blindness associated with damage around the occipital pole to more subtle deficits in particular aspects of vision (e.g., an inability to recognize faces) associated with damage to parts of the temporal and parietal lobes. And, electrophysiological studies in non-human primates had characterized response properties of individual neurons within visual cortex, providing insight into both how those neurons process specific visual features (e.g., lines) and how those neurons are organized into functional units (e.g., ocular dominance columns). Given this considerable body of prior knowledge, some of the very first fMRI studies examined the brain response to simple visual stimuli, so that data collected using this new technique could be compared to known anatomy.

In 1992, Kwong and colleagues sought to measure basic visual responses using the simplest possible technique: comparing brain signals measured during a period of darkness (about a minute) contrasted with those measured during a similar interval containing a bright, flashing light

(Figure 1). This direct comparison between two conditions became known as a “contrast.” Compared to modern studies, much less was known about how changes in neuronal activity lead to signals measurable using MRI. Kwong and colleagues used two distinct techniques, one of which estimated changes in signal thought to be associated with blood flow; this signal was more similar to that used in earlier PET and MRI studies, but is no longer commonly used for fMRI. The other technique was sensitive to changes in blood oxygenation—building on then-recent work by Seiji Ogawa and others—and thus measured the same signal as present-day fMRI studies. As expected, both of these techniques revealed broad activation in the occipital lobe, consistent with neuronal activity in primary visual cortex. Thus, this early study provided proof-of-concept evidence that fMRI could elicit similar information about visual function as prior techniques, albeit at much coarser temporal and spatial resolution.

1998

Within a few years of the first studies, advances in experimental design and computational analysis techniques allowed researchers to construct much more sophisticated maps of the visual system. Instead of presenting simple

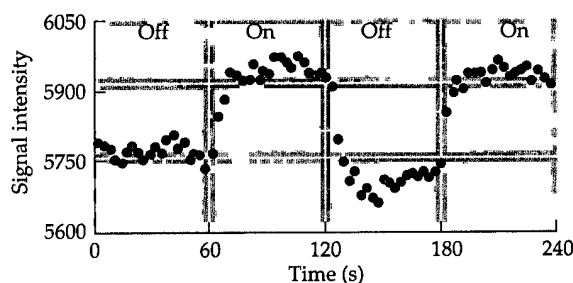
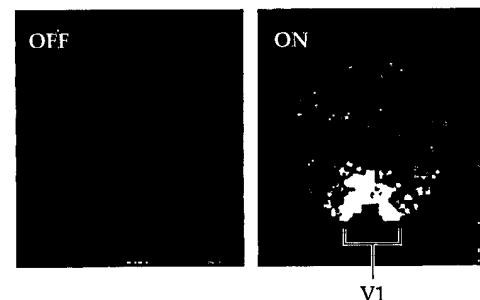


Figure 1 Using fMRI to examine the brain response to simple visual stimuli. Within the posterior occipital lobe, consistent with the location of primary visual cortex (V1; the back of the brain is at the rear of the image), there was increased activation to flashing



visual patterns compared to darkness. This article shows its data in terms of changes in the fMRI hemodynamic response over time, here normalized into arbitrary units. (After Kwong et al., 1992.)

BOX 1.1 (continued)

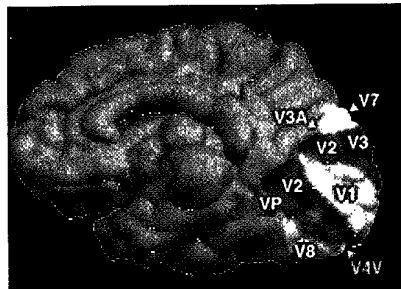


Figure 2 Using fMRI to map the functional organization of visual cortical regions. By using fMRI retinotopy, researchers could construct maps of brain function that are specific to an individual subject's anatomy, here shown on a medial view of a 3D-rendered brain. Each label (e.g., "V1," "V2") refers to a specific functionally defined region within visual cortex. (From Tootell et al., 1998.)

flashing stimuli that filled the visual field, researchers selectively targeted particular parts of visual space (e.g., the upper right of visual fixation). By controlling the spatial pattern of those stimuli—such as by first flashing stimuli at the center of visual space and then flashing progressively more eccentric stimuli—neuronal activity could be driven in a predictable manner. Moreover, new analysis techniques took advantage of known features of the visual system: different regions within the visual cortex contain local organization according to their spatial features (e.g., eccentricity), and thus local discontinuities in the brain response to those features provided a guide to the boundaries between visual regions. This approach, which came to be known as fMRI retinotopy, could be used to create maps of visual cortex that were specific to an individual participant's functional anatomy (**Figure 2**). Retinotopy has remained an important tool for fMRI, particularly for identifying functional regions in individual subjects for subsequent studies of how those regions are modulated by attention, reward, or other processes.

2006

While contrasts in the amount of activation within a region provide evidence for localization of function—and remain an important aspect of fMRI research

today—advances in fMRI research go beyond simple contrasts between conditions, often integrating data from

more complex patterns of activation. A 2006 paper by Peelen and colleagues used an experimental design relatively similar to that of the 1998 study just described, but adopted a different approach to analysis. They examined the pattern of activation amplitude evoked by different perceptual stimuli across voxels (the small cubes that are elemental components of MR images; **Figure 3A**) within each of several

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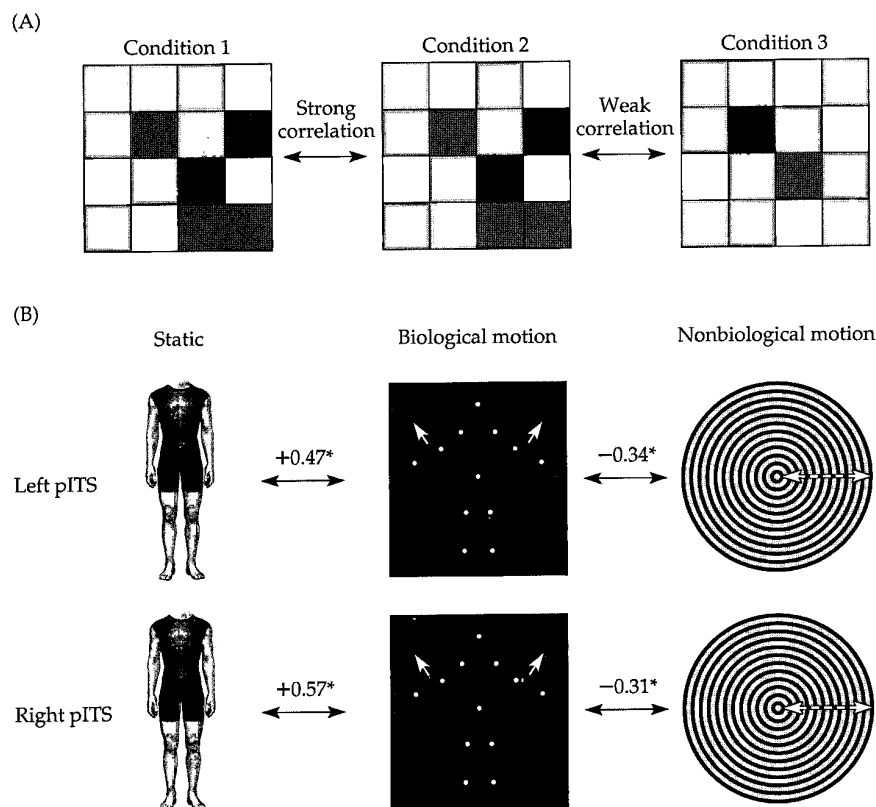


Figure 3 Extending the understanding of biological motion perception by analyzing patterns of brain activation. (A) The researchers measured patterns of activation across voxels and examined whether different pairs of perceptual categories evoked similar or different patterns, as seen in this schematic example. (B) Within the lateral occipital-temporal cortex—here, the posterior inferior temporal sulcus (pITS)—the pattern of activation evoked by biological motion (center) was generally similar to that evoked by perception of static biological stimuli (left) but was negatively correlated with that of nonbiological motion (right). Data are presented as statistical correlations between experimental conditions, such that each number shows the correlation statistic and asterisks indicate statistical significance. Conclusions about brain function are indicated without showing the brain or the hemodynamic timecourse. (After Peelen et al., 2006.)

patients with severe epilepsy to help localize the source of their seizures. However, the electrical and magnetic activity generated inside the brain can be measured from outside the skull using techniques known as **electroencephalography (EEG)** and **magnetoencephalography (MEG)**. These electromagnetic recording methods measure rapid changes in electrical potentials and magnetic flux, making them particularly valuable for studying the timing of brain processes. Because of these differences in what the various techniques measure—and thus in their strengths and weaknesses—neuroscientists match their measurement technique to their specific research goals.

Manipulation techniques evaluate how a change in the brain itself leads to changes in its function. Sometimes the manipulation is unintended, such as a change in the brain due to an injury or stroke. An early and landmark result was reported by the French physician Paul Broca regarding his examination of a single patient named Leborgne. This patient was effectively unable to speak, being able only to repeat the syllable “tan” in response to prompting. At Leborgne’s autopsy in 1861, Broca demonstrated that the patient had damage to the brain that was largely restricted to the inferior frontal lobe in the left hemisphere. This demonstration provided conclusive evidence that language-production abilities are localized, at least in part, to the area of the brain that now bears Broca’s name. During the following decades, many other nineteenth-century researchers created lesions in animals to test whether a brain region must be intact for expression of a behavior. Today, many laboratories collect data on groups of individuals who share damage to some brain region (e.g., people with lesions in their prefrontal cortex), so that researchers can evaluate which functions are impaired by damage to that region.

Lesion studies have unquestionable value in providing compelling evidence that specific brain regions are *necessary* for specific functions. Yet lesion studies also pose challenges. It is often difficult to find human patients with isolated damage. Many trauma or stroke patients have diffuse damage, and their lesions may encompass multiple functional brain regions. One way to overcome this problem is to *create* lesions in a specific region, so that the researcher can control the spatial extent of the damage. Such experimental introduction of permanent lesions is limited to animal models, for obvious reasons. However, the temporary interruption of function within a brain region is possible using **transcranial magnetic stimulation (TMS)**, which can be used in human subjects to disrupt neuronal activity for periods of seconds to minutes. Further description of TMS, including how it is used in parallel with fMRI, will be provided in Chapter 13.

Interpreting deficits in function caused by brain damage, whether permanent from a lesion or temporary from TMS, can be problematic. The fact that damage to area X impairs behavior Y indicates necessity (i.e., the use of area X is necessary for behavior Y to occur), but not sufficiency (i.e., other areas may also contribute to the expression of that behavior). In an oft-cited analogy, damage to any one of several parts of a radio—the speakers, or the tuner, or even the power switch—will lead to its inability to play music. However, you could not damage one of those parts and then correctly claim that the damage to that one part knocked out the “music-playing area” of the radio. As an interconnected part of a complex system, a given brain region may contribute to more than one psychological process or behavior, and a given process or behavior may be supported by multiple brain regions, especially when considering the relatively diffuse effects of many sorts of brain damage. The effects of a lesion also change over time. As the brain heals, an injured region may once again become able to support processing, or other regions may change their processing to compensate for the damage. Furthermore, even a small lesion might result in damage to adjacent fiber pathways, impairing

electroencephalography (EEG) The measurement of the electrical potential of the brain, usually through electrodes placed on the surface of the scalp.

magnetoencephalography (MEG) A noninvasive functional neuroimaging technique that measures very small changes in magnetic fields caused by the electrical activity of neurons, with potentially high spatial and temporal resolution.

transcranial magnetic stimulation (TMS) A technique for temporarily stimulating a brain region to disrupt its function. TMS uses an electromagnetic coil placed close to the scalp; when current passes through the coil, it generates a magnetic field in the nearby brain tissue, producing localized electric currents.

the connections between two distant regions. It is therefore critical to evaluate the effects of many different lesion locations and to track the effects of those lesions over time.

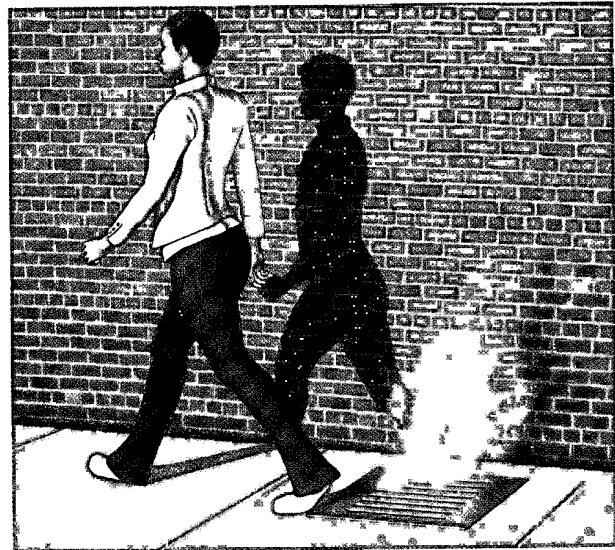
Brain function can also be manipulated through changes in brain chemistry. Neurons throughout the brain have receptors that are sensitive to particular neurotransmitters, such as dopamine or serotonin. Drugs or foods may change the concentrations of these neurotransmitters, or alter the ability of the brain to process them. Some chemical changes modify the activities of specific types of neurons, while others alter how a wide variety of neurons fire and communicate. Drug studies are powerful, in that they allow the investigation of large-scale brain systems that are not associated with simple lesions; they are also clinically relevant, in that many drugs have well-understood effects on brain disorders (e.g., Parkinson's disease and drugs that manipulate the availability of the neurotransmitter dopamine). A central disadvantage, however, is the difficulty in identifying the functions of specific brain regions following the systemic application of a drug. If the motor skills of a patient with Parkinson's disease improve after administration of a drug that supplies dopamine to the brain, that improvement could be due to better functioning in the midbrain, the basal ganglia, the prefrontal cortex, or any number of regions responsive to dopamine. In addition, many drug manipulations have relatively slow time courses, with functional changes that can take place over weeks, thus precluding inferences about short-term cognitive processes.

In conclusion, fMRI is one tool among many available to neuroscientists. It is a noninvasive measurement technique that can provide information about brain metabolism in human subjects, which has made it very popular for studies of cognitive functions. Measurement techniques like fMRI provide one direction of inference: experimental manipulation of behavior (or perception or thought) leads to changes in brain function. Manipulation techniques provide the other direction of inference by revealing how natural variation or experimental manipulation of brain function lead to changes in behavior. Thus these two classes of techniques have complementary uses, and their combination provides a powerful approach to the study of human cognition. However, this complementarity is not absolute, as will be discussed later in this chapter. Chapter 13 provides more detailed information about the many techniques used to study brain function, with a focus on those techniques that are used in conjunction with fMRI.

Key concept: Contrast

Any imaging technique, from X-rays to fMRI, can be evaluated by four simple criteria: what quantity does it measure, how sensitively can it measure that quantity, how precisely in space does it measure that quantity, and how often can it make the measurement? Consider the simple imaging system formed by the sun, you, and a wall. If you stand between the bright sun and the wall, your shadow appears. For opaque objects, such as a person, the shadow will be very dark compared with the wall around it. However, if the sun's rays pass through something insubstantial, like a cloud of steam or a sheer curtain, the shadow will be relatively lighter (**Figure 1.4**). In this imaging system, the quantity being measured is the number of photons of sunlight that strike the wall, and that measurement indicates the degree to which the intervening object absorbs photons. By comparing the shadows cast by different objects, like a person or a cloud of steam, one can estimate the optical opacity of the objects. Here, the difference between dark and light shadows on a wall indicates the opacity (i.e., light transmittance) of the

Figure 1.4 Images and contrast. In this very simple imaging system formed by the sun, two objects (the person and the steam), and a wall, the opacity of the objects can be estimated by the darkness of the shadows that are cast. This imaging system shows contrast based on opacity to visible light.



object being imaged, with dark areas indicating opaque objects and brighter areas indicating transparency. In fact, this simple system captures the essence of the familiar X-ray technique.

The difference between the lightest and darkest shadows is a measure of the contrast available in our imaging system. If the imaging technique is sensitive to small gradations in the quantity being measured, the resulting image will have good contrast and will enable us to discriminate between objects with only slight differences. Contrast, however, is not "all or nothing." Because no imaging method is perfect, there will always be some amount of variation in the measured signal. For example, a plane passing overhead could momentarily block the sun and change the intensity of your shadow on the wall. Thus, it is typical to express contrast in relation to the background variation, or noise, and to discuss the results in terms of the contrast-to-noise ratio (CNR). We will explore this topic in more detail in later chapters.

Depending on the pulse sequence used by the scanner, images can be created that differentiate between low and high proton densities, gray matter and white matter, fluid and tissue, or the direction of water diffusion in tissue. Thus, the quantity being measured is different for each of these image types. In this context, contrast has another, specialized meaning that may be initially confusing. **Figure 1.5** shows images that have contrast based on the intrinsic tissue properties T_1 and T_2 . We will describe these tissue properties

contrast (1) The intensity difference between different quantities being measured by an imaging system. (2) The physical quantity being measured (e.g., T_1 contrast). (3) A statistical comparison of the activation evoked by two (or more) experimental conditions, in order to test a research hypothesis.

contrast-to-noise ratio (CNR) The magnitude of the intensity difference between different quantities divided by the variability in their measurements.

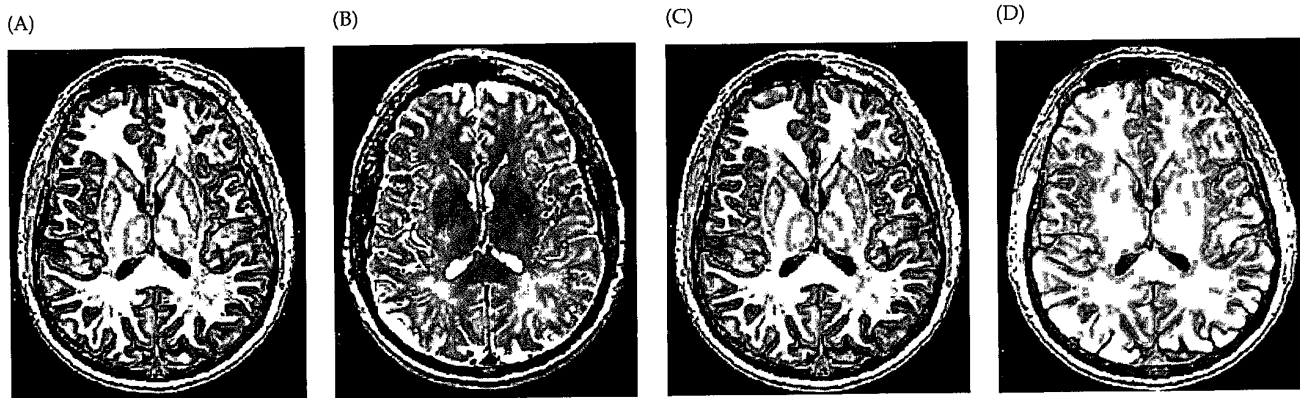


Figure 1.5 Contrast and contrast-to-noise ratios in MRIs. The images in (A) and (B) are sensitive to two different contrast types (T_1 and T_2 , respectively). Although much of the same brain structure is present in both images, the relative intensities of different tissue types are very different. The images in (C) and (D) are of the same contrast type but have

different contrast-to-noise ratios. (C) An image with a very high contrast-to-noise ratio; significant detail can be seen. (D) An image with a lower contrast-to-noise ratio. Some distinctions, such as the boundary between gray and white matter, are difficult to identify.

and how these different types of images are created in Chapters 3 through 5. In T_1 -contrast images, the contrast between light and dark is a measure of the relative difference in the T_1 property of the tissues, which causes fluid to appear as black, gray matter to appear as dark gray, and white matter to appear as light gray. In T_2 -contrast images, the contrast between light and dark measures a different tissue property, and in these images gray matter is light, white matter is dark, and fluid is very bright.

To map brain function, researchers must create images that distinguish between active and non-active areas of the brain (**Figure 1.6**). These images

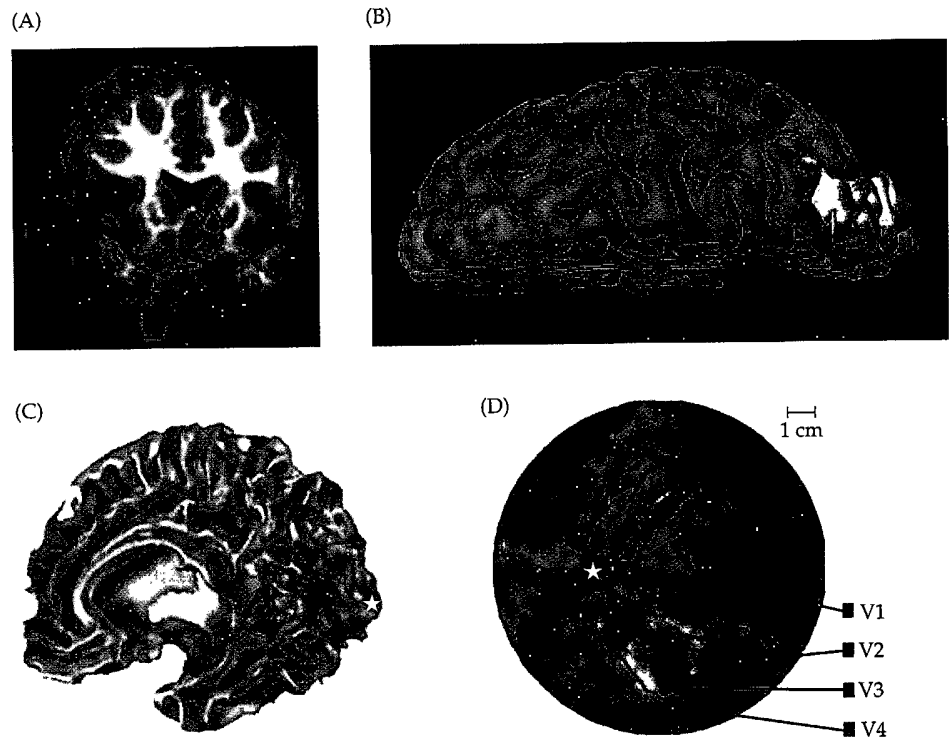


Figure 1.6 Functional maps of the brain produced by fMRI. Even though fMRI data are collected as a series of two-dimensional slices, areas of activation can be visualized in complex, three-dimensional figures. (A) Shown are areas of activation to the expectation of a monetary reward; the red area in the center of the image is the ventral striatum, an area important for reward and learning. (B) This image shows a reconstruction of the right hemisphere of the brain based on a high-resolution anatomical image. Areas of statistically significant activation are indicated in color. The experiment involved a visual search task, and thus areas in the occipital and parietal lobes (at right) are highly active. (C) The cerebral cortex consists of a thin (about 5 mm) sheet of gray matter; its bumps and valleys are visible as the gyri and sulci of the brain. Shown on the reconstructed three-dimensional brain are areas important for visual processing, as identified using fMRI. Different parts of the visual system have been color-coded based on how they responded to stimuli presented to different parts of the retina. (D) This information can be used to identify regions of visual cortex with distinct properties; shown here are the boundaries of four early regions (V1–V4), overlaid on a flattened view of the cortex. For reference, the part of the visual cortex most sensitive to information presented at the center of gaze (i.e., the fovea) is indicated with a star on both images. (C,D courtesy of Dr. Greg Appelbaum, Duke University.)

rely on functional contrast. In PET studies, functional contrast is based on the number of emitted radioactive decay particles. For PET researchers to say that one area of the brain is more active than another, there must be a statistically significant difference between the numbers of particles emitted by each of those regions. In fMRI studies, functional contrast is usually based on the total amount of deoxygenated hemoglobin in the blood, as will be discussed in detail in Chapter 7. Whether a region is classified as active or inactive depends on the change in the amount of deoxygenated hemoglobin measured in that region. We emphasize that the contrast-to-noise ratio, whether anatomical or functional, depends on both the *amount of signal change* and the *variability in the signal*. An image may have high contrast-to-noise despite small absolute intensity differences, if there is very little variability within each property being measured.

Key concept: Resolution

The ability to distinguish different locations within an image is known as spatial resolution. Imagine a digital satellite photograph of a college campus and the surrounding countryside, such as that shown by Google Maps and similar programs. When the photograph covers many miles of terrain, even a large structure such as an enormous athletic stadium might appear as a single dot. But if you zoom in so that the photograph covers only a single street block, then much more detail can be appreciated; now you can see smaller buildings, walkways, and automobiles. In a digital photograph of a scene, the smallest elements that can be resolved are known as pixels, or picture elements. So, in a satellite photograph showing a broad area of countryside, each pixel might represent several hundred yards, while a photograph of a street block may contain pixels representing areas a few feet wide. Similarly, magnetic resonance (MR) images may be able to resolve relatively coarse or fine elements. Since all MR images sample the brain in three dimensions, the basic sampling units of MRI are known as voxels, or volume elements. As voxel size decreases, the ability to identify fine structure in a brain image improves (**Figure 1.7**). In principle, the voxel size in MRI can be made arbitrarily small; high-spatial-resolution images of rodent brains can have voxels less than 0.05 mm on a side. But, as described in Chapter 7, the total signal recovered from a voxel is proportional to its size, and voxels that are too small may have insufficient signal to create high-quality images. In structural MRI of the human brain, voxels are often about 1 mm in each dimension, while voxels collected in fMRI experiments are typically about 3 mm in each dimension.

Although structural MR images are considered to be static representations of the brain, fMRI is inherently dynamic, in that it measures changes in brain activity over time. The rate at which a technique acquires images, or its sampling rate, constrains its temporal resolution. In a typical fMRI study, the sampling rate is often one brain volume every second or two. This is much faster than PET studies, which measure changes in brain metabolism over intervals of a few minutes to many tens of minutes, but much slower than techniques measuring electrical activity directly. Our ability to detect changes in brain function over time is limited not only by the sampling rate, but also by the sluggishness of the physiological changes that we seek to measure. Most fMRI studies measure changes in blood oxygenation, which resolve over periods of a few seconds to a few tens of seconds. Even if we sample the brain very rapidly (say, several times per second), the hemodynamic changes may occur too slowly for us to make inferences about the more rapid neuronal activity.

functional contrast A type of contrast that provides information about a physiological correlate of brain function, such as changes in blood oxygenation.

spatial resolution The ability to distinguish changes in an image (or map) across different spatial locations.

pixel A two-dimensional picture element.

voxel A three-dimensional volume element.

sampling rate The frequency in time with which a measurement is made.

temporal resolution The ability to distinguish changes in a signal (or map) across time.

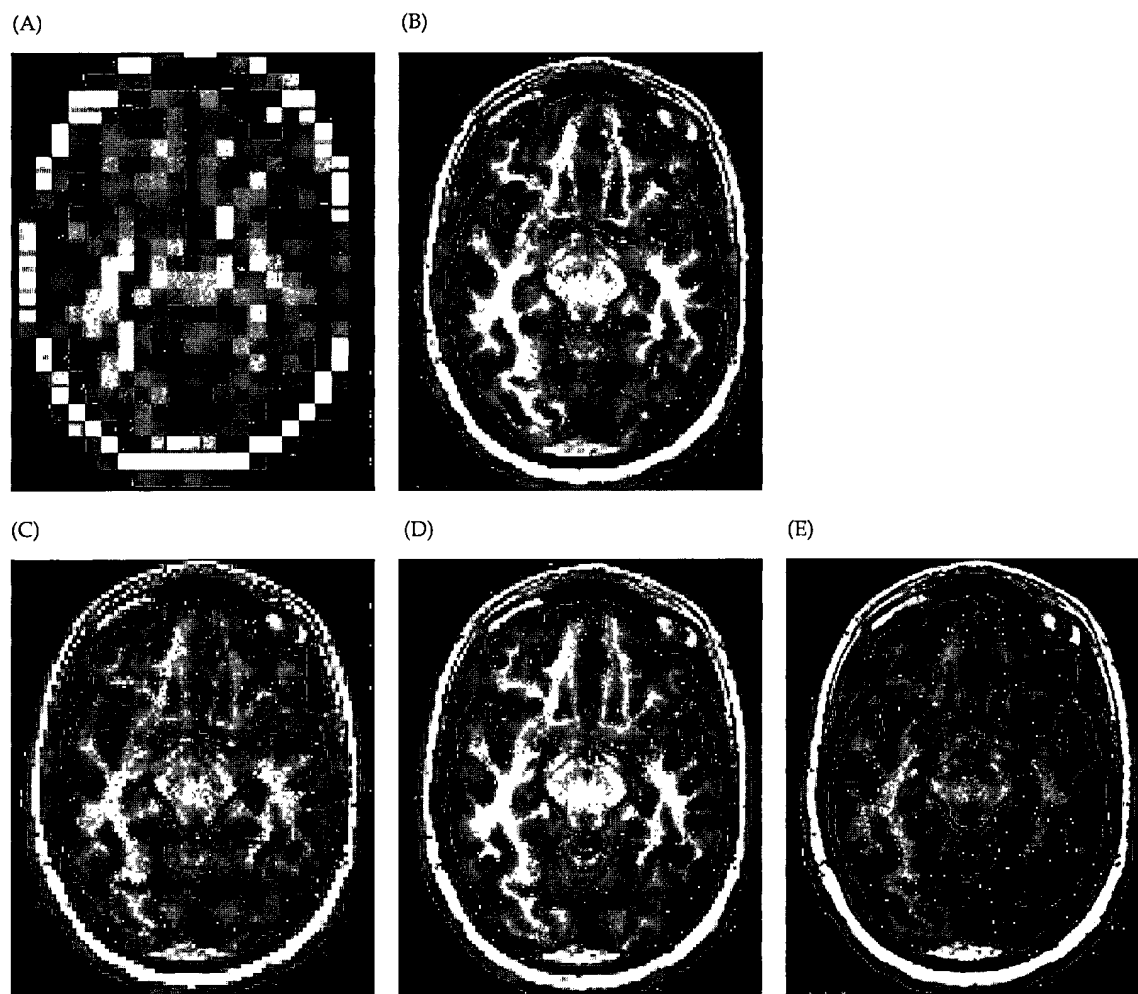


Figure 1.7 The human brain at different spatial resolutions. Spatial resolution refers to the ability to resolve small differences in an image. In general, we can define spatial resolution based on the size of the elements (i.e., voxels) used to construct the image. The images shown here present the same brain sampled at five different element sizes: 8 mm (A); 4 mm (B); 2 mm (C); 1.5 mm (D); and 1 mm (E). Note that the gray-white structure is well represented in the latter three images, all of which were produced using element sizes that were less than half the typical gray matter thickness of 5 mm.

Together, spatial and temporal resolutions have been used to describe a “technique space” that shows how different experimental methods provide potentially complementary information about brain function (**Figure 1.8**). The canonical example of complementarity combines the measurement of brain hemodynamics by fMRI with the measurement of brain electrophysiology by EEG. Because fMRI has very good spatial resolution (millimeters) and EEG has very good temporal resolution (milliseconds), it is argued that combining them would apply the best aspects of each to a single research question. While seductive, this argument conceals a deeper issue that is introduced here and discussed in more detail in Chapter 13. Spatiotemporal graphs like the one in **Figure 1.8** suggest that some fundamental quantity (i.e., “brain

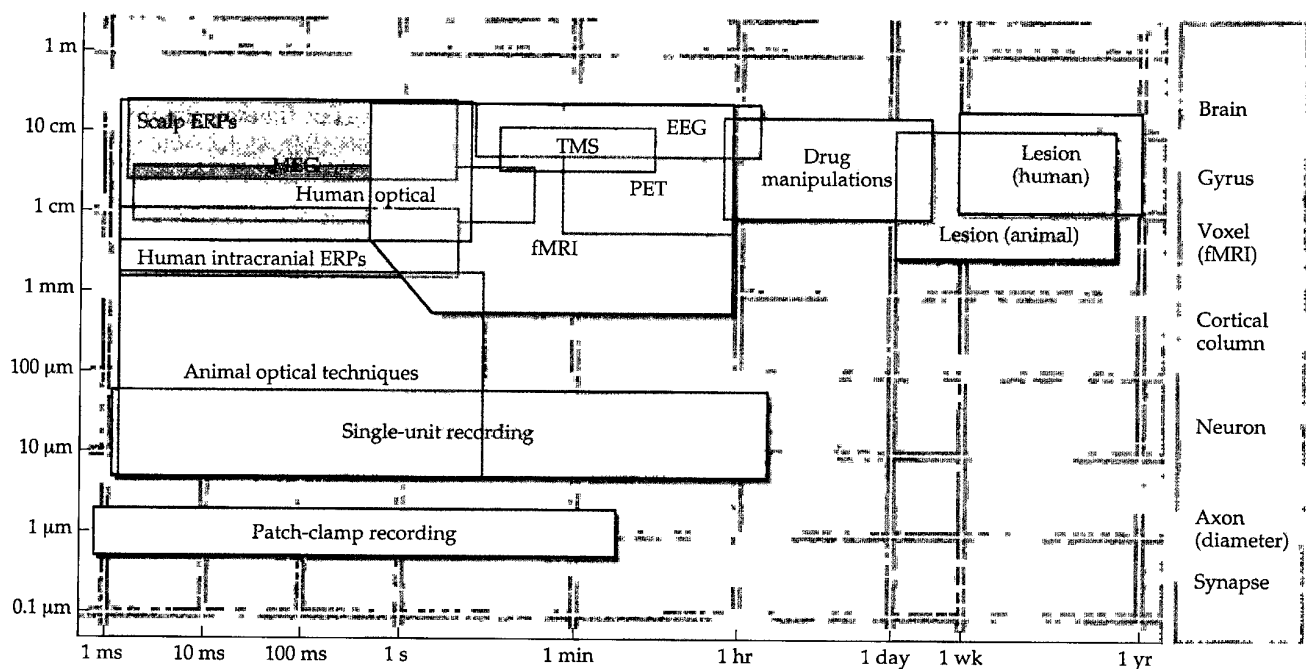


Figure 1.8 Neuroscience techniques differ in temporal resolution (x-axis) and spatial resolution (y-axis). Functional MRI (bright yellow) provides a good balance of spatial and temporal resolution, and thus is appropriate for a wide range of experimental questions. However, other approaches, including electrophysiology, lesion studies, and drug manipulations, can provide important complementary information. ERPs, event-related potentials; MEG, magnetoencephalography; TMS, transcranial magnetic stimulation; EEG, electroencephalography; PET, positron emission tomography.

activity") is a single variable that can be measured at different scales of time and space. However, no such fundamental quantity exists. Rather, each of the techniques measures a different aspect of brain physiology. These measures do not always correlate among themselves and, more importantly, they do not always allow similar sorts of inferences about mental processes. Also, some techniques are appropriate for some subject populations but not others (e.g., animals vs. humans, adults vs. children). Even though spatial and temporal resolutions are important properties for comparing different techniques, they are insufficient and even misleading when exclusively used to decide whether a technique is appropriate for answering a specific research question.

Instead, the value of a technique is determined by its **functional resolution**, or ability to map physiological variation to mental processes or behaviors. All of the above properties—contrast, spatial resolution, and temporal resolution—contribute to functional resolution. Yet, other factors are also critical. The brain property being measured determines, in several ways, how well one can localize function in the brain. The changes in blood oxygenation measured by fMRI reflect the local vascular structure of the brain. In EEG studies using extracranial recording, the local vascular structure has little effect on activity, but the orientation and temporal synchrony of the active neurons have an enormous effect. So even though a given task might evoke significant activity in a particular brain region using one measurement technique,

functional resolution The ability to map measured physiological variation to underlying mental processes or behaviors.

it might not evoke activity in the same region when a different technique is used. Thus, while fMRI allows researchers to draw the broadest set of inferences about the functioning of the intact human brain, in our view it also has many important limitations. We will return to this theme—the limitations of fMRI—throughout the book.

Thought Question

As fMRI has become better established, researchers have become increasingly interested in applying it to clinical questions. What properties of fMRI might make it particularly well (or poorly) suited for studying disorders like schizophrenia, autism, or addiction?

History of fMRI

The scientific developments leading to modern fMRI can be divided into five main phases:

- Basic physics work in the 1920s to 1940s set forth the idea that atomic nuclei have magnetic properties and that these properties can be manipulated experimentally.
- Seminal studies reported by two laboratories in 1946 described the phenomenon of nuclear magnetic resonance (NMR) in solids, ushering in several decades of nonbiological studies.
- The first biological MR images were created in the 1970s after advances in image acquisition methods.
- During the 1980s, MRI became clinically prevalent, and structural scanning of the brain was commonplace. Around that same time, it was discovered that oxygenation levels influenced the magnetic properties of blood.
- In the early 1990s, the discovery that changes in blood oxygenation could influence MR images ushered in a new era of functional studies of the brain.

We provide in this section a brief overview of the history of MRI; we will discuss specific physical principles in more detail in subsequent chapters.

Early studies of magnetic resonance

The beginnings of MRI can be traced to the work of the Austrian physicist Wolfgang Pauli in the 1920s. Noting anomalies in the electromagnetic spectra emitted by excited atoms, Pauli postulated that atomic nuclei had two properties, called spin and magnetic moment, which could only take discrete values (or quanta). As an analogy, think of atomic nuclei as continually spinning tops. Pauli's suggestion in rough terms was that these tops could spin only at certain frequencies and exert only particular magnetic forces. At that time, nuclear properties were poorly understood—indeed, the neutron itself would not be identified until 1932, in a report by the English physicist James Chadwick—and Pauli's conjecture would not be tested for more than a decade.

In an early technique for investigating whether different atomic nuclei spin at discrete frequencies, a gaseous beam of a single element was passed through a strong static magnetic field before hitting a detector plate. A static magnetic field is one whose intensity does not change over space or time. If the spin

frequencies of atomic nuclei could only take a number of discrete quantum states, then the static magnetic field would split the beam into some finite number of smaller beamlets before the beamlets hit the detector. On the other hand, if the spin frequencies of the atomic nuclei in the beam could take a continuous range of possible values, then there would be a similarly continuous distribution of intensity on the detector. The beams did split into discrete beamlets, as predicted by quantum theory. These results proved that the spin frequency of an atomic nucleus can take only one of a number of discrete values.

The spin frequency values for particular atomic nuclei, however, remained to be measured. In 1933, the American physicist Isidor Rabi (Figure 1.9A) modified the molecular beam technique so that he could measure the spin frequencies of hydrogen and alkali metals. But Rabi felt that this beam technique was inelegant and sought a better method. The Dutch physicist Cornelis Gorter visited Rabi's laboratory in 1937 and described his experiments with oscillating magnetic fields. Stimulated by this discussion, Rabi realized that if the frequency of the oscillating magnetic field could be matched to the spin frequency of the atomic nucleus, then the nucleus would absorb energy from the field. This is called magnetic resonance. To understand this idea, consider

oscillating magnetic field A magnetic field whose intensity changes periodically over time. Most such fields used in MRI oscillate at the frequency range of radio waves (megahertz, or MHz), and as such they are often called radiofrequency fields.

magnetic resonance The absorption of energy from a magnetic field that oscillates at a particular frequency.

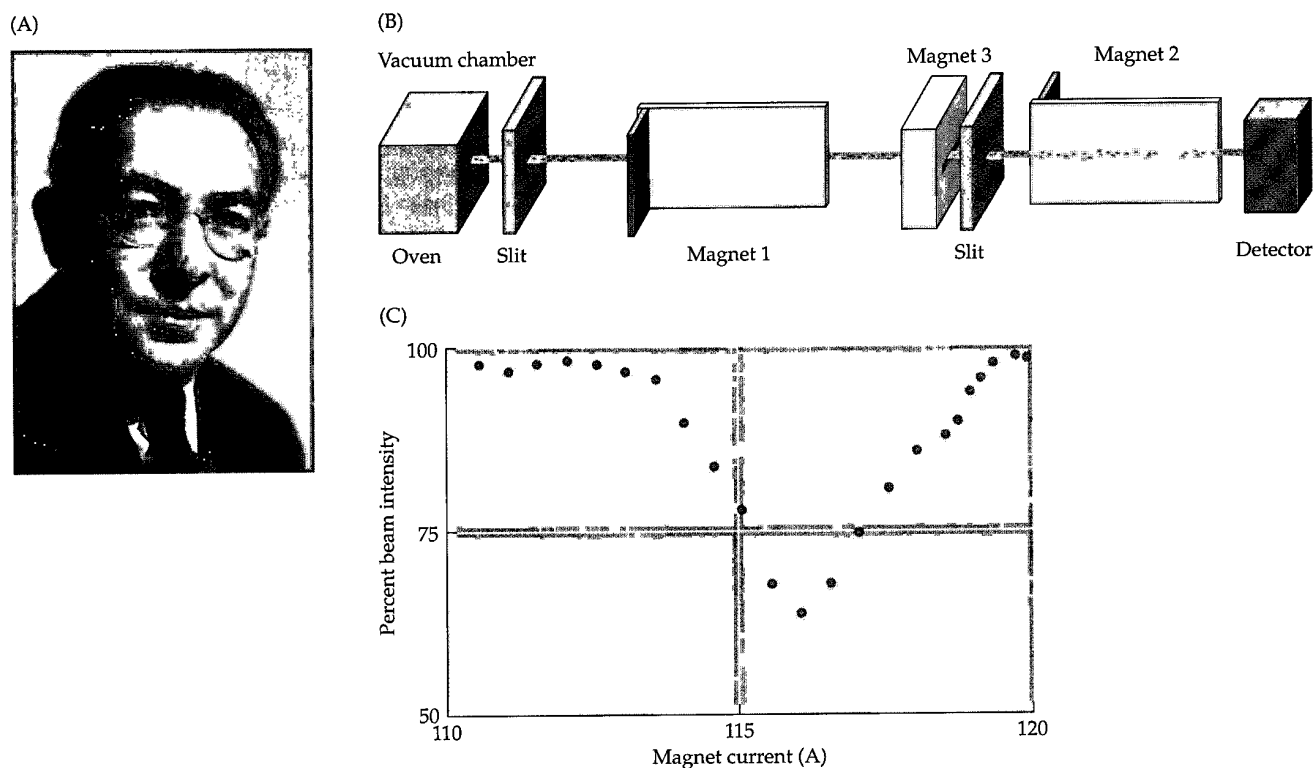


Figure 1.9 The determination of the magnetic moment of the lithium nucleus by Isidor Rabi (A). The beam technique devised by Rabi involved passing a beam of gaseous nuclei through several magnetic fields (B). The key innovation introduced by Rabi was an oscillating electromagnetic field (Magnet 3). If the oscillation rate was equal to the resonant frequency of an atomic nucleus (at the current strength of the static magnetic field), the spin of the atomic nuclei would change and then

the subsequent magnetic field (Magnet 2) would deflect the nuclei away from the detector. (C) Data from Rabi's experiment, in which he kept the oscillation rate fixed and changed the current in the static field magnet to modify its magnetic field strength. He found a sharp reduction in beam intensity at about 116 amperes, allowing him to calculate the spin properties of the lithium nucleus. (A © The Nobel Foundation.)

resonant frequency The frequency of oscillation that provides maximum energy transfer to the system.

(A)



(B)

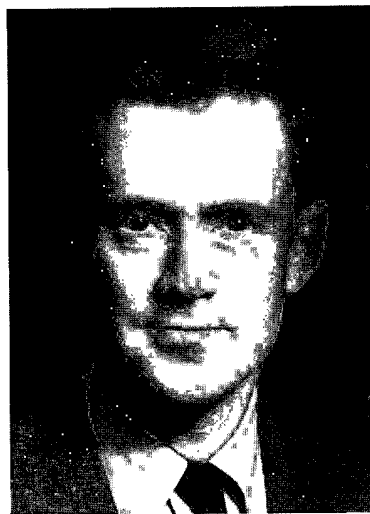


Figure 1.10 Felix Bloch (A) and Edward Purcell (B) shared the 1952 Nobel Prize in Physics for their simultaneous but separate discoveries of magnetic resonance in bulk matter. (© The Nobel Foundation.)

the analogy of a swing set, to which we will return in Chapter 3. If your friend is sitting on a swingset, you can help her swing back and forth by pushing her. A single hard push will have only a limited effect. But by pushing her gently, regularly, and in synchrony with her swinging, at each cycle she will swing a little higher. The frequency of pushing that has the most effect is known as the **resonant frequency**. Energy can be given to atomic nuclei in the same way, by a large number of small energy transfers from a magnetic field that oscillates at the resonant frequency of the nucleus.

This idea catalyzed work in Rabi's lab, and just days after Gorter's visit, the molecular beam technique was modified to include an oscillating magnetic field. Rabi recognized that the resonant frequency needed for the oscillating field would depend on the strength of the static magnetic field, just as the speed at which someone swings would depend on the strength of the gravitational field. So, he held the frequency of the oscillating field constant and changed the strength of the static field by adjusting the current in the magnet. (Note that this approach is the opposite of that used by modern MR scanners, which keep the static field constant and vary the oscillating field to examine different atomic nuclei.) As the strength of the main field approached the resonant frequency of the lithium atoms in the beam, the atoms in the beam were deflected away from the detector (**Figure 1.9B,C**). This experiment represented the first demonstration of nuclear magnetic resonance effects, for which Rabi received the Nobel Prize in Physics in 1944.

NMR in bulk matter: Bloch and Purcell

During the early 1940s, much basic research in physics stopped because the top physicists of the day were working on military applications, such as the development of the atomic bomb and the improvement of radar and counter-radar measures. When the war ended in 1945, two physicists, Felix Bloch at Stanford (**Figure 1.10A**) and Edward Purcell at MIT/Harvard (**Figure 1.10B**), resumed their independent investigations of magnetic resonance in bulk matter (i.e., normal solid substances). The previous magnetic resonance experiments by Rabi and others had used beam methods that required purified gases. For magnetic resonance to become practical as a measurement technique, it would need to be applicable to normal substances, not just laboratory creations like beams of atoms.

On December 13, 1945, Purcell and his colleagues began their first experiment to demonstrate magnetic resonance in bulk matter. Borrowing a strong magnet that had originally been used for astronomical research, they placed paraffin wax into the center of the magnetic field. They reasoned that if they matched the resonance frequency of the wax to the oscillating magnetic field, the wax would absorb energy. This, in turn, would change the wax's electrical conductance, which could be detected by a simple circuit. Like Rabi, they recognized that the resonant frequency of the wax would depend on the static magnetic field strength, so they changed the current flowing through the coils of their electromagnet to change its field strength. Despite their careful planning, when they adjusted the current in the magnetic field no resonance was found!

At first, Purcell and his colleagues suspected that they had not left their wax sample in the magnetic field for a long enough time before initiating their experiment. Scientists had previously theorized that it took some time before atomic nuclei became aligned with an external magnetic field, a concept known as **relaxation time** (discussed in detail in Chapter 3). However, Purcell and his colleagues did not know the relaxation time for the atomic nuclei in their sample—it could be as short as a few seconds or as long as a few years. On

the chance that relaxation was a very slow process, they placed the wax in the magnetic field for several hours before repeating their experiment two days later. But, even with this presaturation period, the second experiment failed. At the predicted current level of the magnet, there was no resonance effect. Finally, before ending the experiment, the researchers decided to test all possible magnetic field strengths, so they increased the magnet current to the maximum level and then slowly decreased it. To their surprise, they found a clear resonance effect at a near maximum value that was much higher than the one they had predicted; subsequent investigation revealed that they had simply miscalculated how much current was necessary to generate the appropriate magnetic field. Their discovery was reported in *Physical Review* in January 1946.

Almost simultaneously, Bloch and his colleagues at Stanford were also attempting to measure resonance effects in bulk matter, although they were using a very different apparatus. They placed a sample of water in a brass box between the poles of a strong magnet whose field strength they could manipulate. An adjacent transmitter coil sent electromagnetic energy into the sample, while a separate detector coil was used to measure changes in the energy absorbed by the water (as emitted back to the environment). As in the MIT/Harvard experiment, the sample was presaturated for 24 hours in the magnetic field to ensure that relaxation would take place. These long presaturation periods turned out to be excessive; for substances like paraffin and water, relaxation times are only a few seconds rather than many hours.

In a striking parallel to Purcell's experiment, Bloch's group also detected magnetic resonance effects in their water sample. They labeled this phenomenon nuclear induction, and reported their findings in *Physical Review* two weeks after Purcell's report. Nuclear induction, now known as nuclear magnetic resonance (NMR), forms the basis for all modern MRI techniques, and all MR scanners share the basic design principles of Bloch's simple apparatus: a strong static magnetic field, and coils that transmit and measure electromagnetic energy. (Modern MRI scanners also include a third type of electromagnetic coil that generates spatial gradients in the magnetic field.) Purcell and Bloch were awarded a joint Nobel Prize in Physics in 1952 for their independent contributions to the discovery of nuclear magnetic resonance.

Although the discoveries that enabled NMR were made by physicists, the first applications of the new technique came from chemistry. By the early 1950s, the Varian Associates Company, with Felix Bloch's assistance, had patented the basic ideas for using NMR to do chemical analyses of samples. NMR soon proved to be a useful technique for understanding the chemical composition of a homogeneous substance, sharing both theory and methodology with modern MRI spectroscopy. Despite (or perhaps because of) the considerable commercial success of NMR in fields like geology and organic chemistry, the primary uses of the new technique would remain chemical rather than biological for more than two decades.

transmitter coil An electromagnetic coil that generates an oscillating magnetic field at the resonant frequency of atomic nuclei within a sample.

detector coil An electromagnetic coil that measures energy emitted back to the environment after its initial absorption by the sample.

nuclear induction The initial term for nuclear magnetic resonance effects, as labeled by Bloch and colleagues.

nuclear magnetic resonance (NMR) The measurable changes in magnetic properties of atomic nuclei induced by the application of an oscillating magnetic field at the resonant frequency of the nuclei.

The earliest MR images

By the late 1960s, NMR measurements had revealed differences between water molecules depending on whether or not they were within biological tissues. Specifically, the atomic nuclei contained in water molecules in biological tissues were constrained in their diffusion and orientation when compared with water in other states, and these differences could be identified using NMR. The American physician Raymond Damadian hypothesized that similar differences might be observed between cancerous and non-cancerous cells; if so, NMR could become an extremely useful method for identifying cancerous

image In MR imaging, a visual description of how one or more properties of the atomic nuclei within a sample vary over space.

spatial gradients (G) A magnetic field whose strength varies systematically over space. Note that because a given spatial location only experiences one magnetic field, which represents the sum of all fields present, spatial gradients in MRI act to change the effective strength of the main magnetic field over space.

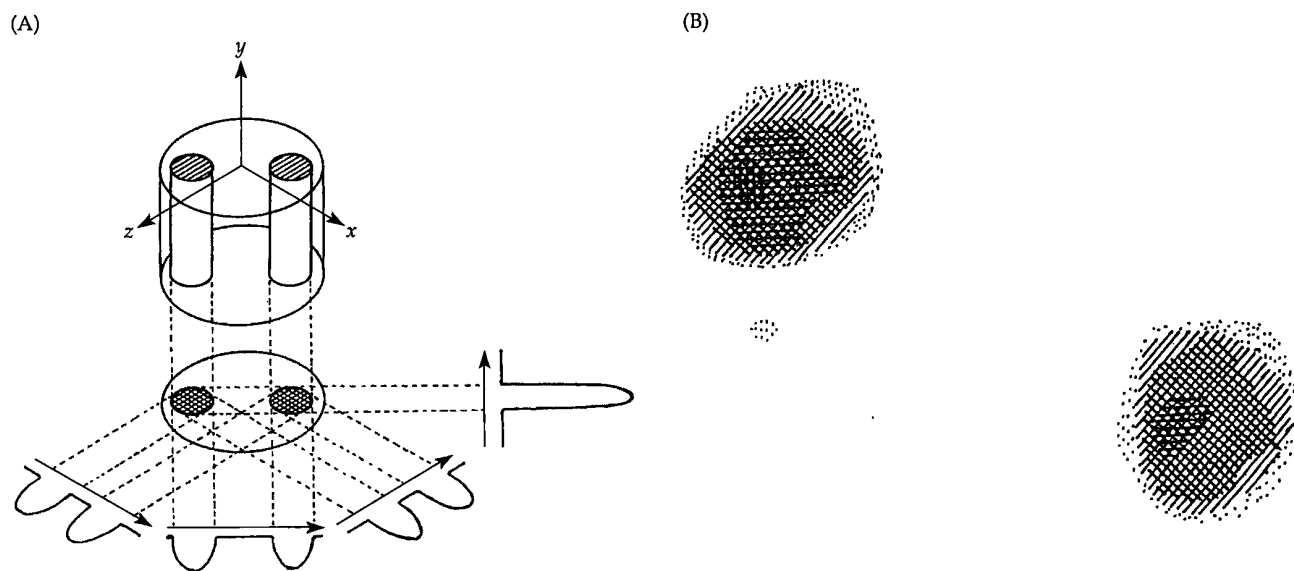
Figure 1.11 The first MR image.

(A) The physicist Paul Lauterbur used a series of spatial gradients to take a succession of measurements of a beaker containing two water-filled test tubes. (B) Data collected under each gradient provided different information about the object. By combining these data using projection methods, Lauterbur was able to reconstruct the spatial organization of the object. The resulting picture was the first magnetic resonance image. The use of spatial variation in the magnetic field set the stage for modern MR imaging. (From Lauterbur, 1973.)

tissue. Damadian tested this hypothesis in tissue samples taken from rats, and found that relaxation times were much longer in the cancerous tissue than in healthy tissue; this result was published in *Science* in 1971. Yet, this early momentum soon faded, as imaging techniques of that time did not provide a reliable advance over existing techniques. The primary influence of Damadian's results turned out to be conceptual, in that for the first time there was a path toward a medical application for NMR. Still to come, however, was the advance that led to the explosion of interest in magnetic resonance: its use to form spatial images.

In a formal sense, an image provides information about how one or more quantities vary over space. As examples, standard photographs are images of the intensity (and frequency) of visible light, and X-ray films are images indicating the density of intervening matter. The first NMR studies did not create images because they measured the total energy absorbed and emitted by the entire sample. In fact, in order to improve the interpretation of their results, early NMR researchers strove to remove spatial information from their samples by making them as homogeneous as possible. Without image formation, NMR remained a relatively obscure curiosity.

Having seen the results of NMR experiments like those conducted by Damadian, the American physicist Paul Lauterbur recognized that NMR had considerable potential for biological and physical applications if a method for image formation could be developed. In 1972, Lauterbur had a novel idea: if the strength of the magnetic field were varied over space, the resonant frequencies of protons at different field locations would also vary. By measuring how much energy was emitted at different frequencies, one could identify how many protons were present at each spatial location. This idea, of inducing spatial gradients (often symbolized as G) in the magnetic field, proved to be the fundamental insight that led to the creation of MR images. Lauterbur also realized that a single gradient could only provide information about one spatial dimension; to recover two-dimensional structure, it would be necessary to use gradients at different orientations. By acquiring data using four gradients in succession, each turned 45° from its predecessor, Lauterbur created an image of a pair of water-filled test tubes (**Figure 1.11**). This picture, which was reported in *Nature* in 1973, was the very first MR image.



Lauterbur's method, though revolutionary, was inefficient because it essentially required a succession of one-dimensional projections through the object and then combined the data into a two-dimensional image; this projection technique was similar to medical imaging using computerized tomography. Not only was there considerable redundancy in the data that were collected, but also the approach was quite time-consuming due to the need for many separate acquisitions. A more efficient technique, known eventually as echo-planar imaging (EPI), was proposed by the British physicist Peter Mansfield in 1976. EPI allowed the collection of data from an entire image slice at one time, by sending one electromagnetic pulse from a transmitter coil and then introducing rapidly changing magnetic field gradients while recording the MR signal. The resulting complex MR signal could be reconstructed into an image using Fourier analysis techniques, as will be discussed in Chapter 4. EPI reduced the time needed to collect a single image from minutes down to fractions of a second, which greatly improved the feasibility of clinical imaging. Concepts derived from EPI underlie the most important approaches to MRI today, and they have been particularly important for fMRI studies due to the need for fast imaging to measure changes in brain function. Lauterbur and Mansfield were jointly awarded the Nobel Prize in Physiology or Medicine in 2003 for their independent contributions to image formation using MRI (**Box 1.2**).

Even with the theoretical underpinnings of MRI largely in place, significant engineering problems were yet to be solved. In 1977, the first human NMR scanner was created by Damadian's Fonar Corporation, and christened "Indomitable" (**Figure 1.12A**). At that time it was difficult to create a strong, homogeneous magnetic field in a scanner large enough to fit an adult human. The magnetic field in Damadian's scanner was weak (0.05 T), and was homogeneous only within a small volume at its center. Data could be acquired from a single small part of the body at one time, whereupon the subject would have to be moved so that another part of the body could be located at the

echo-planar imaging (EPI) A technique that allows collection of an entire two-dimensional image by changing spatial gradients rapidly following a single excitation pulse from a transmitter coil.

(A)



(B)

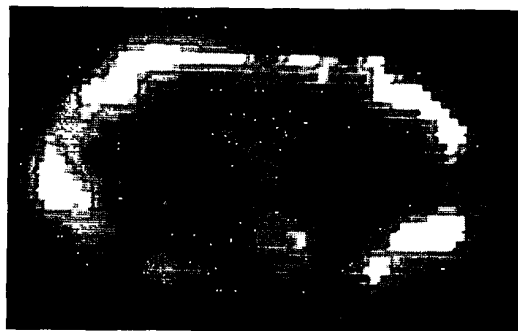


Figure 1.12 The first MR image of the human body. Raymond Damadian and colleagues constructed an early large-bore MRI scanner, which they named "Indomitable." (A) Larry Minkoff, a postdoctoral fellow in the laboratory, was the subject from whom the data were recorded. (B) The resulting image of Minkoff's chest, while primitive by modern standards, shows the heart, lungs, and surrounding musculature. (A courtesy of Fonar Corporation; B from Damadian et al., 1977.)

BOX 1.2 A Nobel Prize for MRI

On October 9, 2003, readers of the *Washington Post* were greeted with a full-page advertisement titled "This Year's Nobel Prize in Medicine—The Shameful Wrong that Must Be Righted" (Figure 1). Similar advertisements appeared the next day in the *New York Times* and *Los Angeles Times*, all of which were paid for by the Fonar Corporation, founded by Raymond Damadian. They protested the announcement, made earlier that week, of the 2003 Nobel Prize in Physiology or Medicine, which was granted to the physicists Paul Lauterbur and Peter Mansfield (Figure 2).

The core argument in these advertisements rested on simple and compelling facts. Damadian demonstrated, in a 1971 *Science* article, that NMR measurements could distinguish between healthy and cancerous tissues. In 1972 he filed the first patent for an MR scanner. And, as recounted in this chapter, he and his colleagues built the first MR scanner that could create images of the entire human body. The rules of the Nobel Prizes allow the granting of each year's prize to up to three individuals, so the selection committee could have chosen to include Damadian (or any other MR pioneer) in addition to Lauterbur and Mansfield. As summarized in the advertisements: "The Nobel Prize Committee to Physiology or Medicine chose to award the prize, not to the medical doctor/research scientist who made the breakthrough discovery on which all MRI technology is based, but to two scientists who later made technological improvements based on his discovery."

These arguments have intuitive force. Damadian clearly played *some* role in the development and popularization of MRI—consider the photograph and scan in Figure 1.12, which document the first human MRI scanning. So, why was he not recognized

(A)



(B)

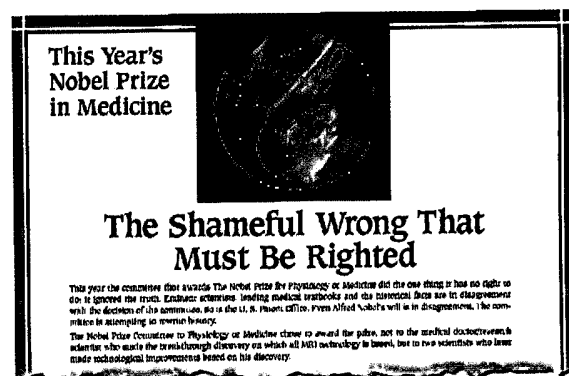


Figure 1 Raymond Damadian, an early pioneer in MRI research, was passed over for the 2003 Nobel Prize. (A) Damadian was one of the first scientists to suggest a biological use for Nuclear Magnetic Resonance measurement and was also the first to build an MR scanner suitable for whole-body human imaging. (B) Upon learning of the Nobel Committee's award, Damadian's MRI company, Fonar Corporation, sponsored advertisements in several leading newspapers to protest Damadian's exclusion. Part of one of those advertisements is shown here. (A courtesy of Fonar Corporation.)

by the Nobel Institute? The deliberations of its selection committee are sealed for 50 years, so the reasons underlying their decision will not be known for some time. Yet, some clues are evident in the press release, which is available for public review on the Nobel Prize website.

The press release begins by describing the basic principle of nuclear magnetic resonance—that atoms in a magnetic field will absorb, and later emit, energy if that energy is delivered at a particular resonant frequency (see Chapter 3). This discovery was the basis of the 1952 Nobel Prize in Physics. Lauterbur and Mansfield were cited not in Physics, but in Physiology or Medicine, because of the applications of their work to medical imaging. Lauterbur's citation reads in part: "introduction of gradients in the magnetic field made it possible to create two-dimensional images." Similarly, "Mansfield utilized gradients in the magnetic field in order to more precisely show differences in

the resonance." The comments of the Nobel Institute indicate that the selection committee recognized Lauterbur and Mansfield for their development of methods for creating *images*. Each made a seminal discovery in the use of magnetic gradients. Lauterbur provided the first demonstration of image creation, and Mansfield developed the method of rapid image collection that is still used today.

What about the claim in the *Washington Post* advertisement that Damadian "made the breakthrough discovery on which all MRI technology is based"? While the first MRI scanner was indeed built by Damadian, it used a point-by-point collection process that was inefficient and quickly abandoned. Nor was the proposal that relaxation properties would predict cancer supported by subsequent work. In short, neither current MRI practice nor its applications follow directly from Damadian's discoveries. Yet there may have been indirect influences. As Damadian and his supporters have argued, the

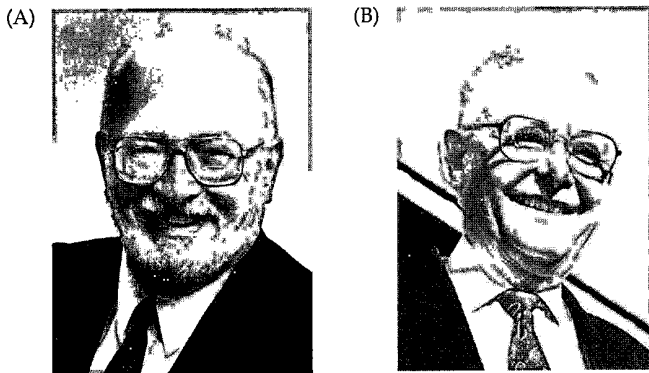
BOX 1.2 (continued)

Figure 2 Paul Lauterbur (A) and Peter Mansfield (B) shared the 2003 Nobel Prize in Physiology or Medicine for contributions to the development of MRI. Lauterbur was cited for his introduction of magnetic field gradients, which changed the spin frequencies of atomic nuclei over space and thus allowed the recovery of spatial information. Mansfield was recognized for his development of echo-planar imaging methods, which allowed rapid collection of images. (A © The Nobel Foundation; B courtesy of Lisa Gilligan, University of Nottingham.)

idea of using NMR to study biological systems may have sparked Lauterbur to introduce magnetic gradients. (Lauterbur was familiar with Damadian's work and had seen a variant of the experiment, but dismissed it as a significant influence on his own work.)

Rarely has the selection of a Nobel award been as immediately controversial as the 2003 Prize in Physiology or Medicine. Some experts in the field believed that a deserving pioneer and inventor was unfairly excluded from recognition, perhaps because of his perceived brashness and ambition, or because he developed his inventions into a business. Others argued that the Nobel Committee chose correctly by emphasizing image processing rather than inventions. Consider these questions: Will a future Nobel Prize be awarded for functional MRI? And, if so, what discoveries will be recognized?

scanner center. This approach did not take advantage of the gradient methods developed by Lauterbur and Mansfield, but instead required the collection of each voxel as an independent NMR volume.

The first attempt to collect an NMR image using Indomitable failed, perhaps due to problems with adjusting the transmitter/receiver coil system to fit the large subject, Damadian himself (see the 1996 book by Mattson and Simon for a complete narrative). After several months of additional adjustment and preparation, they were ready for another attempt. On the morning of July 3, 1977, Larry Minkoff, a postdoctoral fellow in Damadian's laboratory, entered the scanner. They slowly collected data from one voxel at a time, changing Minkoff's position slightly after each acquisition. Each voxel in the image took more than 2 minutes to acquire, and the complete image of 106 voxels took nearly 4 hours. At the end of the marathon session, the researchers' patience was rewarded with a single slice through Minkoff's torso (**Figure 1.12B**). Within 2 years, Damadian's group and other laboratories had created multiple images of the abdomen, upper torso, and head. In addition, researchers and clinicians could now acquire images of the brain from new orientations that were previously difficult to acquire using X-ray-based scanning. Soon, the new mathematical approaches developed by Mansfield allowed collection of images in seconds or minutes rather than hours.

With image formation using NMR a reality, many medical applications rapidly became evident. In the late 1970s, computerized tomography (CT) imaging was commonly used to generate high-resolution images of the human body. CT uses a beam of X-rays that rotates around the body part of interest;

ionizing radiation. Electromagnetic radiation with sufficient energy to break chemical bonds.

the X-ray absorption at each angle is measured, and the resulting projections are combined to form a single picture representing one plane through the tissue. While CT is still commonly used, it does require concentrated X-ray exposure. Because NMR images could provide similar information without X-ray exposure, there was substantial interest in NMR as a potential diagnostic tool. Around the same time, the term "nuclear magnetic resonance" fell into disfavor. It was abandoned partly because of the negative health connotations of the word "nuclear," and this change was justified because NMR does not use ionizing radiation. Also, administrative pressures within hospitals pushed for the separation of MR scanning (which would soon become a dominant technique within radiology departments) from nuclear medicine departments. As a result of these factors, by the early 1980s, NMR became MRI: magnetic resonance imaging.

Growth of MRI

As we have seen, MRI has three primary advantages over other imaging techniques:

- It creates very high spatial resolution images that can visualize a variety of tissues (e.g., both bones and soft tissues in the same image).
- It does not require ionizing radiation, as do X-rays or CT scans.
- It can obtain images in any plane through the body.

However, the capital costs involved in setting up an MRI center were high, especially for hospitals facing budgetary constraints in the recessionary economy of the early 1980s. A typical MRI scanner could cost up to \$1 million to purchase, with maintenance, personnel, and supply contracts adding hundreds of thousands of dollars annually. Many hospitals had recently invested heavily in expensive CT scanners and were loath to commit more funds to an unproven technique. In addition, at that time the U.S. Food and Drug Administration had approved MRI for research purposes only, so insurance companies would not reimburse hospitals for performing the procedure. Nevertheless, there was enough appreciation of the potential of MRI that a number of companies, including Fonar, General Electric (GE), Philips, Siemens, and Varian, began developing MRI scanners for clinical use.

One consequence of the investment in MRI research and development was a substantial increase in the strength of the magnetic fields used in scanners. The standard resistive magnets were replaced by superconducting magnets, which had fewer limitations on field strength and homogeneity. However, expensive cooling agents were needed to maintain superconductivity. GE created the first commercial 1.5-T scanners for the human body in 1982, and began shipping them to hospitals shortly thereafter. While other companies focused on low- to medium-field scanners (0.1 to 1.0 T), GE's emphasis on high-field MRI resulted in considerable market success. As MRI grew, 1.5-T scanners would be the standard workhorses for clinical imaging for more than two decades. Only in the mid-2000s would 1.5-T scanners be replaced by newer 3.0-T scanners, many of which also had updated electronics and multichannel recording capabilities.

A key contributor to the rapid growth of MRI was financial: in 1985, MRI scanning was approved as a standard component of clinical care, opening the door for MRI scans to be prescribed by physicians and billed to insurance companies and Medicare. Rather than having to subsidize the enormous costs of scanners that were used for research purposes, hospitals now saw scanners

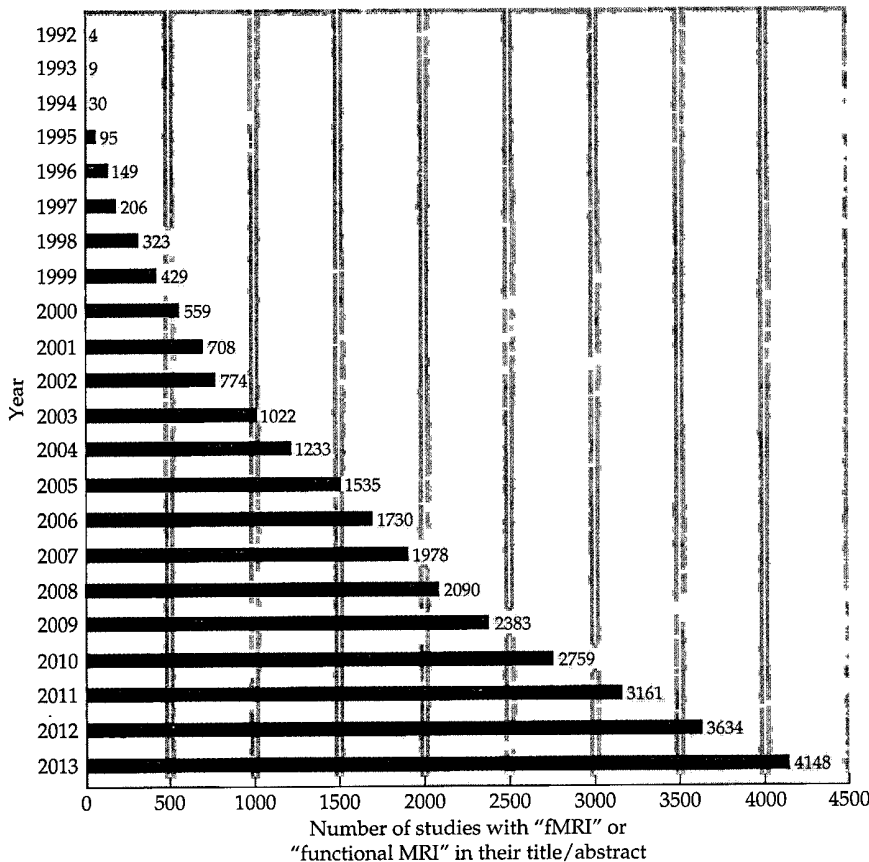


Figure 1.13 Growth in fMRI research. Plotted for each year is the number of PubMed-indexed studies with the terms "fMRI" or "functional MRI" in their titles or abstracts. This necessarily underestimates the true volume of fMRI research, but it does provide a rough indicator of the change in the pace of research since the first studies were published in 1992. Of these nearly 29,000 fMRI studies, about 85% have been published since the first edition of this textbook in 2004, and some 55% have been published since the second edition in 2009.

as sources of profit, which could be obtained by both billing for the scans and attracting new patients. Over the following decade, thousands of MRI scanners were installed in North America alone, making structural MRI one of the most common diagnostic imaging procedures. This prevalence of MRI scanners was a necessary precondition for the explosion of interest in fMRI. Without the clinical need for MRI that led to the installation of many new scanners, the dramatic expansion in fMRI research, graphically illustrated in **Figure 1.13**, would not have occurred. We will return to this history in Chapter 7, which will describe how the clinical MRI scanning of the 1980s led to the functional MRI scanning of the 1990s and thereafter.

Organization of the Textbook

The history of fMRI we have just outlined suggests an incremental progression, beginning with basic physics (and biological principles), through improvements in image collection and hardware design, to the relatively recent onset of fMRI as a neuroscientific tool. The chapters in this textbook recapitulate this progression, by first establishing a foundation based on the physical and biological properties of fMRI and then describing how modern researchers use fMRI to study brain function.

We continue our introduction in Chapter 2, which describes how MRI scanners work. It is important to note that even the most modern MRI scanner

has the same basic components as those used in the experiments described above: a strong static magnetic field to reorient atomic nuclei, weaker but directionally oriented gradient magnetic fields to introduce spatial variation in nuclear magnetic properties, and oscillating radiofrequency fields to induce changes in the energy states of the atomic nuclei. While these three components are integral parts of all MRI scanners, their generators are not the only hardware needed for fMRI. Additional equipment ensures the homogeneity of the magnetic field, measures physiological changes such as heart rate and respiration, presents experimental stimuli, and records participant behavior. Even though MRI presents no dangers to research subjects if conducted correctly, the strong and rapidly changing magnetic fields used in scanning present a set of safety challenges. As safety should be the primary concern for all fMRI experimenters, Chapter 2 also discusses these challenges in detail.

Physical bases of fMRI

From this introduction, we turn to the physics of nuclear magnetic resonance. Chapter 3 introduces the basic principles of magnetic resonance: how atomic nuclei behave in magnetic fields, how the energy states of those nuclei can be changed by applying oscillating magnetic fields at particular frequencies, and how magnetization recovers or decays over time. This discussion leads to the concept of differential relaxation times across tissues, which provide the contrast we measure in MR imaging. The key ideas are first presented in a *conceptual path* that builds from physics principles to MR signal generation in a straightforward and equation-free manner. Then, the same structure is more fully fleshed out in a *quantitative path* that derives the key equations governing MR signal generation.

For “nuclear magnetic resonance” to become “magnetic resonance imaging,” spatial information must be recovered from the raw MR signal. In Chapter 4 we discuss how the introduction of magnetic gradients allows the measurement of signal changes across space. This process is known as image formation. As discussed earlier in this chapter, the development of image formation technology sparked the explosive growth in the use of magnetic resonance for clinical purposes—and subsequently allowed the development of fMRI. We introduce the concept of *k*-space, which provides a useful metric for understanding the relationship between scanning hardware and the measured data.

The final chapter in this section, Chapter 5, links biology and physics by describing the different approaches used by researchers to measure brain structure and function. The extraordinary power of MRI for biological imaging comes from its flexibility: by changing the properties of the gradient and the oscillating magnetic fields over time, images sensitive to many different types of contrast can be obtained. We discuss two major classes of contrasts, and the pulse sequences used to acquire them. Static contrasts provide information about the characteristics of the atomic nuclei at a spatial location, such as their density or tissue type. Motion contrasts provide information about how atomic nuclei, such as those within water molecules in axons or blood vessels, change position over time. We focus on how images can be made sensitive to changes in brain function, which will set the stage for the discoveries outlined in the following sections of the book.

Principles of BOLD fMRI

In Chapters 6 through 8, we introduce the biological underpinnings of fMRI. For neuroimaging to be possible, there must be physiological markers of

brain activity that can be measured. In general, there are two types of markers that are of interest to physiologists. Researchers can measure the *direct consequences of neuronal activity*, such as changes in electrical potentials or chemical gradients, or they can measure the *metabolic correlates of neuronal activity* when they cannot measure the activity itself. Functional MRI relies on the latter approach: it measures blood oxygenation level, which changes in response to the metabolic requirements of active neurons. Chapter 6 describes the metabolic demands of the brain and how these demands are met by vascular function. We also provide a primer for gross brain anatomy, as a background for subsequent discussions of fMRI research.

Even if the metabolic consequences of neuronal activity are understood, there remains the challenge of measuring how metabolic responses change using MRI. The technique that is the basis for nearly all fMRI studies is blood-oxygenation-level-dependent (BOLD) contrast. Chapter 7 begins with a history of the developments that led to the discovery of BOLD contrast in the 1990s. We discuss the early fMRI studies that first demonstrated the feasibility of this technique, and how those studies extended previous neuroimaging work with PET. We return to concepts like spatial and temporal resolution. Although the spatial resolution of fMRI is often claimed to be extremely good, there are many challenges to accurate mapping, notably that of translating between neuronal activity and measurable changes in the vascular system. Conversely, the temporal resolution of fMRI is often claimed to be rather poor, despite approaches to improving temporal resolution, many of which have been used to identify sub-second changes in activity. Throughout the chapter we emphasize how the design choices made by researchers influence these properties.

In Chapter 8, we describe issues related to signal and noise in fMRI data. The central problem in fMRI research is that the signal associated with the fMRI hemodynamic response is very small compared with other sources of variability in the data. We provide definitions of experimental signal and noise and show how different signal-to-noise ratios influence the spatial and temporal patterns of the activity measured by fMRI. We also discuss methods for improving the sensitivity of fMRI, such as improving experimental power through signal averaging, and the elimination of extraneous physiological variability. Many of the experimental procedures used to improve signal-to-noise ratios are described under the general term “preprocessing,” which refers to the steps taken to reduce the variability in the data that is unrelated to the experimental task. Common preprocessing steps include correction for head motion, co-registration of functional images to higher-resolution anatomical images, and normalization of each subject’s brain to match the shape, size, and orientation of a standard reference brain.

Design and analysis of fMRI experiments

In Chapter 9 we discuss the key concepts of experimental design. Most fMRI studies now use rapid event-related designs—that is, they examine changes in the fMRI signal associated with discrete events like the presentation of stimuli or the responses of the participant. Such designs take advantage of some important properties of the fMRI hemodynamic response, which allow the hemodynamic responses associated with an individual event within a sequence to be extracted, independent of the responses to other events. The growth of event-related designs has allowed fMRI to be applied to a broad set of empirical questions; nearly every aspect of human thought or behavior can now be addressed using fMRI. Because of the critical importance

blood-oxygenation-level-dependent (BOLD) contrast The difference in signal on T_2^* -weighted images as a function of the amount of deoxygenated hemoglobin.

of experimental design for all fMRI experiments, we provide guidelines for thinking about good design, and we identify some common problems that can derail otherwise promising experiments.

Experimental design is intimately tied to the analytic approaches available to fMRI researchers. Chapter 10 considers the major statistical tests used for hypothesis testing in fMRI studies. These are discussed within the framework of the general linear model, which guides the analysis of most fMRI studies. While the statistical approaches used in fMRI research share many similarities with those of other domains of science, there are some challenges that are particularly relevant to fMRI. Chief among these is the problem of multiple comparisons: when tens of thousands of voxels are subjected to a statistical test, many effects may appear to be significant merely by chance. We discuss the various approaches to compensating for the multiple comparisons problem, including thresholding, smoothing, cluster analyses, and region-of-interest analyses.

Chapter 11 extends this discussion by introducing advanced methods for data analysis. Our primary focus is on "data-driven" methods. Broadly considered, such a method begins with the observed data, then creates a model for the pattern of brain activation (and thus the types of cognitive processing) that could have generated those data. Common methods include the examination of functional connectivity, the estimation of causal relations between brain regions, and the application of machine learning models that use fMRI data to predict participants' behavior or experiences. Although as yet used in only a minority of fMRI studies, data-driven analyses are becoming an increasingly important aspect of fMRI experimentation.

Applications and future directions

Perhaps nothing testifies to the pace of change in fMRI methodology more than this simple fact: with each new edition of this textbook, topics considered in the previous edition's chapter on advanced fMRI methods have become commonplace in research. The continual development of new methods represents one of the most exciting aspects of the field. New ways to improve spatial resolution, temporal resolution, or signal-to-noise ratio will quickly become adopted throughout the field. In Chapter 12, we take on the difficult task of describing (and predicting) the next generation of fMRI methods.

Chapter 13 describes fMRI in the context of other neuroscientific techniques. Throughout the textbook, we emphasize that fMRI is just one of many techniques available to the neuroscientist. Although it is a flexible and powerful tool, fMRI nevertheless has many limitations that can be addressed by conducting studies that combine fMRI with other techniques. We discuss the theoretical basis for studies that combine multiple approaches, such as using fMRI to identify active brain regions that can then be temporarily disabled using magnetic stimulation, or comparing fMRI data from human and non-human primate subjects.

Finally, Chapter 14 outlines a set of ethical issues underlying fMRI research. Some ethical questions are rooted in practical considerations of MRI scanning; for example, how should researchers deal with the potential discovery of brain abnormalities in seemingly normal subjects? Others are more conceptual: how can we use fMRI to confirm, reject, or change models (or even thinking) of human behavior? Scientists within many disparate fields, from linguistics to marketing, now incorporate fMRI into their research programs. Yet it often remains unclear whether fMRI data are even compatible with these other fields, with some skeptics taking the strong position that fMRI is

irrelevant for modeling human behavior. We end the book by speculating on the future directions for fMRI research.

Summary

Functional magnetic resonance imaging, or fMRI, is one of the most important techniques for understanding the human brain in action. Although fMRI is a relatively new technique, the developments that led to its current stage span nearly a century. Many of the advances that made fMRI possible resulted from basic physics research, and the experimental apparatuses used in these early studies laid the groundwork for modern MRI equipment. Unlike most structural MRI, which measures differences between tissues, most functional MRI studies measure changes in the blood oxygenation of the brain over time. From these changes, researchers make inferences about the underlying neuronal activity and how different brain regions may support different perceptual, motor, or cognitive processes. The strengths of fMRI include: it is noninvasive; it can be used in a wide range of human subject populations; its spatial and temporal resolution are well matched to functional changes in the brain; and it is adaptable to many types of experimental paradigms. While fMRI cannot address every neuroscience question, it provides important measurements of brain function that complement the information obtained from other techniques.

Suggested Readings

Bandettini, P. A. (2012). Twenty years of functional MRI: The science and the stories. *NeuroImage*, 62, 575–588. *This article introduces a special issue of the journal NeuroImage that is devoted to the history of fMRI, through review articles and retrospective pieces from many of the leaders in the field.*

*Bloch, F., Hansen, W. W., and Packard, M. (1946). Nuclear induction. *Phys. Rev.*, 69, 127. *This very short note describes the discovery of nuclear magnetic resonance in solid matter, for which Felix Bloch would share the 1952 Nobel Prize in Physics with Edward Purcell.*

Finger, S. (2000). *Minds behind the Brain*. Oxford University Press, New York. *This very accessible set of short biographies of neuroscience pioneers contains a description of the life and works of Franz Joseph Gall, who founded the discipline of phrenology.*

*Lauterbur, P. C. (1973). Image formation by induced local interactions: examples employing nuclear magnetic resonance. *Nature*, 242: 190–191. *This seminal article describes the first use of magnetic field gradients for the formation of images using magnetic resonance.*

*Mattson, J., and Simon, M. (1996). *The Pioneers of NMR and Magnetic Resonance in Medicine: The Story of MRI*. Dean Books, Jericho, NY. *This encyclopedic text describes in detail the histories of major twentieth-century figures in the discovery of MRI, providing both personal and scientific context for their work.*

*Purcell, E. M., Torrey, H. C., and Pound, R. V. (1945). Resonance absorption by nuclear magnetic moments in a solid. *Phys. Rev.*, 69: 37–38. *Published scant weeks before Bloch's similar report, this article describes the procedures and results from the very first study of nuclear magnetic resonance.*

Purves, D., Cabeza, R., Huettel, S. A., LaBar, K. S., Platt, M. L., and Woldorff, M. (2012). *Principles of Cognitive Neuroscience*. 2nd edition. Sinauer, Sunderland, MA. *This textbook provides a broad introduction to the research methods and key findings in cognitive neuroscience, and thus includes many examples drawn from fMRI research.*

*Indicates a reference that is a suggested reading in the field and is also cited in this chapter.



Refer to the

fMRI

Companion Website at
sites.sinauer.com/fmri3e
 for study questions and
 Web links.

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